

Early Detection of Parkinson through Speech Analysis



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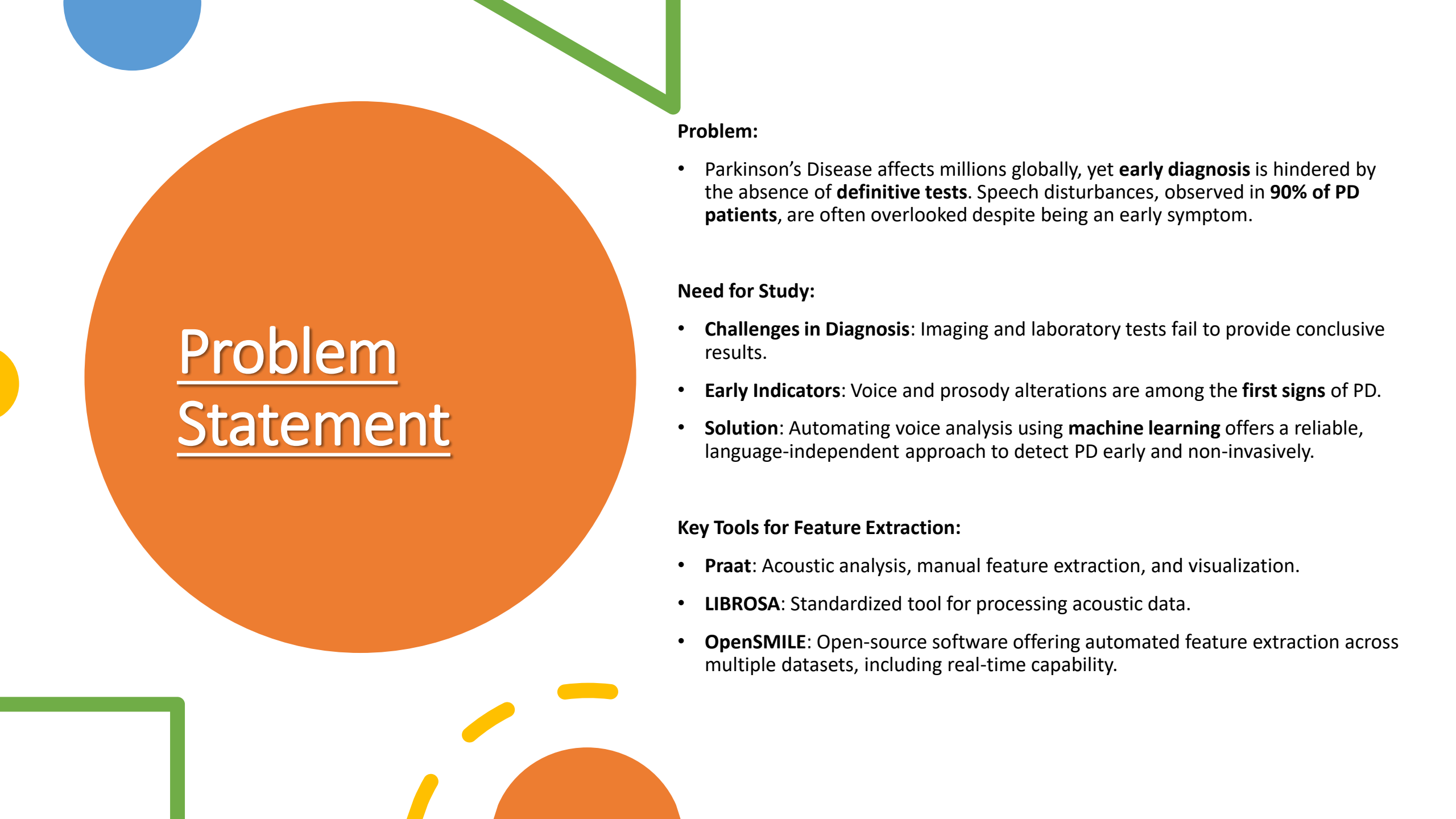
Addanki Naveen

Ram Swarup Bavikati



Parkinson's Disease

- Parkinson's Disease (PD) is a **progressive neurodegenerative disorder** primarily affecting **dopamine-producing neurons**. The disease manifests through motor symptoms such as **tremors, rigidity, bradykinesia**, and **postural instability**, along with non-motor symptoms like **speech impairments**, which are among the earliest indicators.
- Despite its prevalence, diagnosing PD remains challenging due to the lack of **definitive biomarkers** or **reliable imaging tools**. Symptoms are often detected after **significant neuronal loss**, delaying intervention. Early diagnosis is critical for better **disease management** and improved patient outcomes.



Problem Statement

Problem:

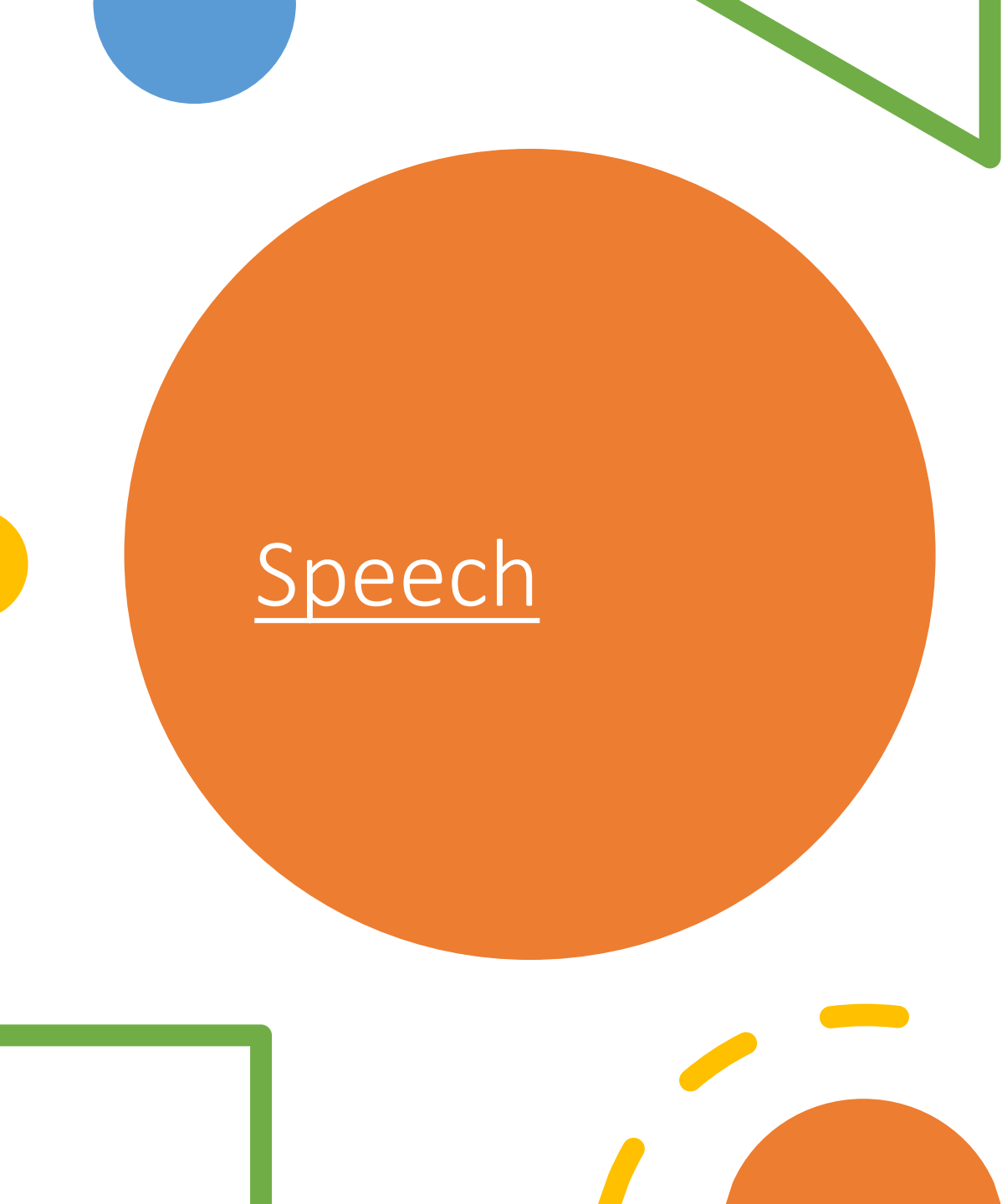
- Parkinson's Disease affects millions globally, yet **early diagnosis** is hindered by the absence of **definitive tests**. Speech disturbances, observed in **90% of PD patients**, are often overlooked despite being an early symptom.

Need for Study:

- **Challenges in Diagnosis:** Imaging and laboratory tests fail to provide conclusive results.
- **Early Indicators:** Voice and prosody alterations are among the **first signs** of PD.
- **Solution:** Automating voice analysis using **machine learning** offers a reliable, language-independent approach to detect PD early and non-invasively.

Key Tools for Feature Extraction:

- **Praat:** Acoustic analysis, manual feature extraction, and visualization.
- **LIBROSA:** Standardized tool for processing acoustic data.
- **OpenSMILE:** Open-source software offering automated feature extraction across multiple datasets, including real-time capability.



Speech

Speech as a Biomarker

Speech provides an excellent biomarker for PD as it is **non-invasive, cost-effective**, and easily monitored. The disease affects multiple components of **speech production**:

- **Phonation Changes:** Reduced loudness, roughness, tremor in voice.
- **Articulation Challenges:** Imprecise consonants, reduced vowel space, slurred speech.
- **Prosodic Features:** Monotone pitch, variable rate, inappropriate pauses.

These changes are **language-independent** and detectable in early stages, making **speech analysis** an effective diagnostic tool.

Speech Disturbances in Parkinson's Disease:

PD significantly impacts communication, with speech disturbances emerging as one of the earliest and most disabling symptoms:

Prevalence: Almost **90% of PD patients** experience hypokinetic speech.

Characteristics:

- Monotonous pitch and low tone.
- Variable speech speed with inappropriate pauses.
- Shortened phrases, articulation decay, and vocal tremors.

These symptoms worsen as the disease progresses, emphasizing the need for diagnostic tools focusing on **speech analysis**

Why OpenSMILE?

OpenSMILE (Open Speech and Music Interpretation by Large-space Extraction) has become the **industry standard** for speech feature extraction, widely used in **research, clinical applications, and emotion recognition**. Its success is rooted in its ability to provide:

- **Reproducible results** backed by active community support and ongoing research validation.
- Access to **extensive feature sets**, enabling both basic and advanced speech analysis.
- Application in **clinical diagnostics**, computational paralinguistics, and real-time processing.

By leveraging OpenSMILE's advanced toolkits, researchers can extract a wide array of **speech and voice quality parameters**, making it a versatile solution for tasks ranging from **emotion detection** to **pathology analysis**.



Feature Sets in OpenSMILE

OpenSMILE provides **flexible feature** sets tailored for diverse use cases. Key sets include:

1. ComParE_2016 (6,373 Features)

Designed for computational paralinguistics challenges, this set is ideal for **detailed speech analysis, emotion recognition, and research requiring extensive features.**

- Includes MFCC 1-14, voicing parameters, and spectral analysis.

2. eGeMAPS v01a/v01b (~88 Features)

Purpose-built for **clinical applications and voice pathology detection**. This compact set emphasizes frequency, energy, and spectral parameters.

- v01b improves upon v01a with enhanced stability and bug fixes.

3. GeMAPS v01a/v01b (62 Features)

Designed for **real-time and mobile-friendly applications**, GeMAPS is the go-to solution for quick voice analysis in resource-constrained environments.

- Features include essential **frequency metrics and minimal spectral descriptors.**

4. emobase (~988 Features)

Specialized for **emotion and affect recognition**, this set includes prosodic and energy-based features for **basic emotion classification.**



OpenSMILE

Feature Processing

OpenSMILE processes speech features at two primary levels, enabling a balance between precision and computational efficiency.

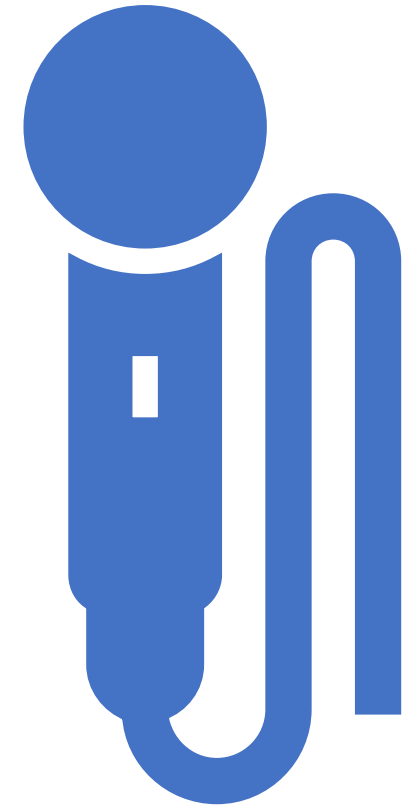
1. Low-Level Descriptors (LLD):

- Conducts **frame-by-frame analysis** to extract core acoustic features such as **fundamental frequency, spectral features, and voice quality**.
- Focuses on granular details **for high-resolution speech characterization**.

2. Functional-Level Descriptors:

- Aggregates statistical and temporal information from LLDs to derive **global descriptors**.
- Computes metrics such as **mean, standard deviation, contour descriptors**, and **temporal features** for a holistic analysis of speech patterns.

This layered approach ensures flexibility, catering to both **real-time applications** and **complex diagnostic tasks**.





Literature Study

Machine learning approaches to identify Parkinson's disease using voice signal features

Parkinson's Disease (PD):

- Second most common age-related neurological disorder.
- Early symptoms often misdiagnosed due to overlap with normal aging or other disorders like essential tremor.

Challenges in Diagnosis:

- No definitive lab biomarkers or imaging techniques available.
- Voice dysfunction, one of the earliest indicators, offers potential for early non-invasive detection.

Dataset Details:

- 195 voice recordings from 31 subjects (23 PD, 8 Healthy Controls).
- Features include measures like fundamental frequency (Fo), jitter, shimmer, and harmonic-to-noise ratio.

Proposed Methodology:

- Utilized Machine Learning (ML) and Deep Learning (DL) models:
 - Support Vector Machine (SVM), Random Forest (RF), K-Nearest Neighbor (KNN), Decision Tree (DT), Multi-Layer Perceptron (MLP).
- Addressed data imbalance using SMOTE (Synthetic Minority Over-sampling Technique).
- Enhanced model performance with hyperparameter tuning via GridSearchCV.

Key Findings:

- MLP achieved the highest accuracy (98.31%) with excellent precision, recall, and F1-score.
- SVM also performed well with 95% accuracy.
- Voice-based analysis proves to be a reliable, cost-effective, and non-invasive diagnostic approach.

Significance:

- Highlights voice analysis as a biomarker for early PD detection.
- Suggests integration into healthcare for accessible, scalable diagnostics.

Dataset

Voice Samples for Patients with Parkinson's Disease and Healthy Controls

[Cite](#)[Download all \(4.18 MB\)](#)[Share](#)[Embed](#)[+ Collect](#)

Dataset posted on 2023-08-03, 20:44 authored by [Fred Prior](#), Tuhin Virmani, Anu Iyer, Linda Larson-Prior, Aaron Kemp, Yasir Rahmatallah, Lakshmi Pillai, Aliyah Glover

The Demographics file contains: Sample ID, participant age, sex and label - HC (Healthy Control) or PwPD (Patient with Parkinson's Disease). The Sample ID is a key that links to the voice recording files PD_AH.zip and HC_AH.zip.

Voice recordings are in .wav format and represent a prolonged enunciation of the vowel /a/. These recordings were made using participant's telephones.

FUNDING

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Give Feedback

Adapted Methodologies

1. Data Collection and Preprocessing:

Dataset:

- Utilized the Voice Samples for Patients with Parkinson's Disease and Healthy Controls dataset containing 81 voice recordings from 81 subjects (40 with PD, 41 healthy controls).
- Features extracted included measures like fundamental frequency (Fo), jitter, shimmer, and harmonic-to-noise ratio.

Preprocessing Steps:

Class Balancing:

- Applied **Synthetic Minority Over-sampling Technique (SMOTE)** to address data imbalance, ensuring equal representation of healthy and PD samples.
- **Definition:** SMOTE generates synthetic samples for the minority class by interpolating between nearest neighbors in the feature space.

Standardization:

- Used StandardScaler to normalize features for consistent data distribution and improved model performance.
- **Definition:** StandardScaler standardizes features by removing the mean and scaling to unit variance.

Adapted Methodologies

2. Feature Selection:

- Applied **SelectKBest** to identify the most relevant features for classification.
- **Definition:** SelectKBest uses statistical tests to rank features by importance, reducing dimensionality and improving model efficiency.
- **Top Features Selected:** MDVP:F0(Hz), MDVP:F0(Hz), MDVP:Shimmer, HNR, PPE, and spread1.

Adapted Methodologies

3. Machine Learning and Deep Learning Models:

Trained and tested various algorithms for classification:

- **Support Vector Machine (SVM):**
 - Separates data using a hyperplane for optimal class distinction.
 - Effective for both linear and non-linear separations using kernels.
- **Random Forest (RF):**
 - Combines multiple decision trees to improve accuracy and reduce overfitting.
- **K-Nearest Neighbor (KNN):**
 - Classifies data points based on the majority class of its nearest neighbors in feature space.
- **Decision Tree (DT):**
 - Splits data into branches based on decision rules from features, making it interpretable and efficient.
- **Multi-Layer Perceptron (MLP):**
 - A type of artificial neural network with one input layer, one or more hidden layers, and an output layer.
 - Used for non-linear problem solving with backpropagation for weight optimization.

Adapted Methodologies

4. Hyperparameter Tuning:

- Used **GridSearchCV** for systematic exploration of hyperparameter combinations.
- **Definition:** GridSearchCV performs an exhaustive search over a specified parameter grid using cross-validation to find the best-performing model configuration.

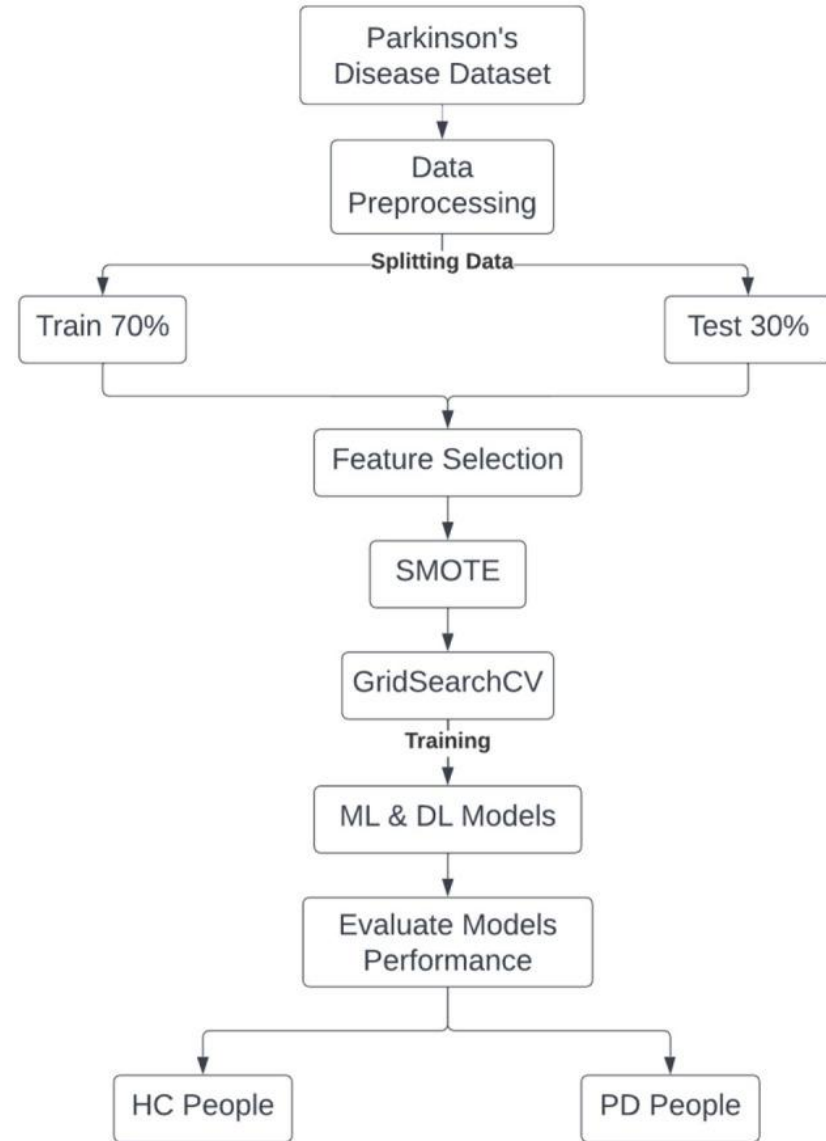
5. Performance Evaluation:

Evaluated model effectiveness using key metrics:

- **Accuracy:** Measures overall correctness of the model.
- **Precision:** Proportion of true positives to total predicted positives.
- **Recall (Sensitivity):** Ability to detect true positive cases.
- **F1-Score:** Harmonic mean of precision and recall, balancing the two metrics.

Confusion matrix used to analyze classification outcomes (True Positive, True Negative, False Positive, False Negative).

Architecture Diagram



Feature Analysis



Low-Level Descriptors (LLDs)

These are basic acoustic features describing the raw characteristics of the voice, such as frequency and amplitude variations.



Prosodic Features

These relate to rhythm, intonation, and stress in speech, often indicative of voice abnormalities.



Voice Quality Features

These describe the phonatory characteristics of the voice, such as noisiness and tonal quality.



Temporal Features

These capture the timing and sequential aspects of speech, including classification



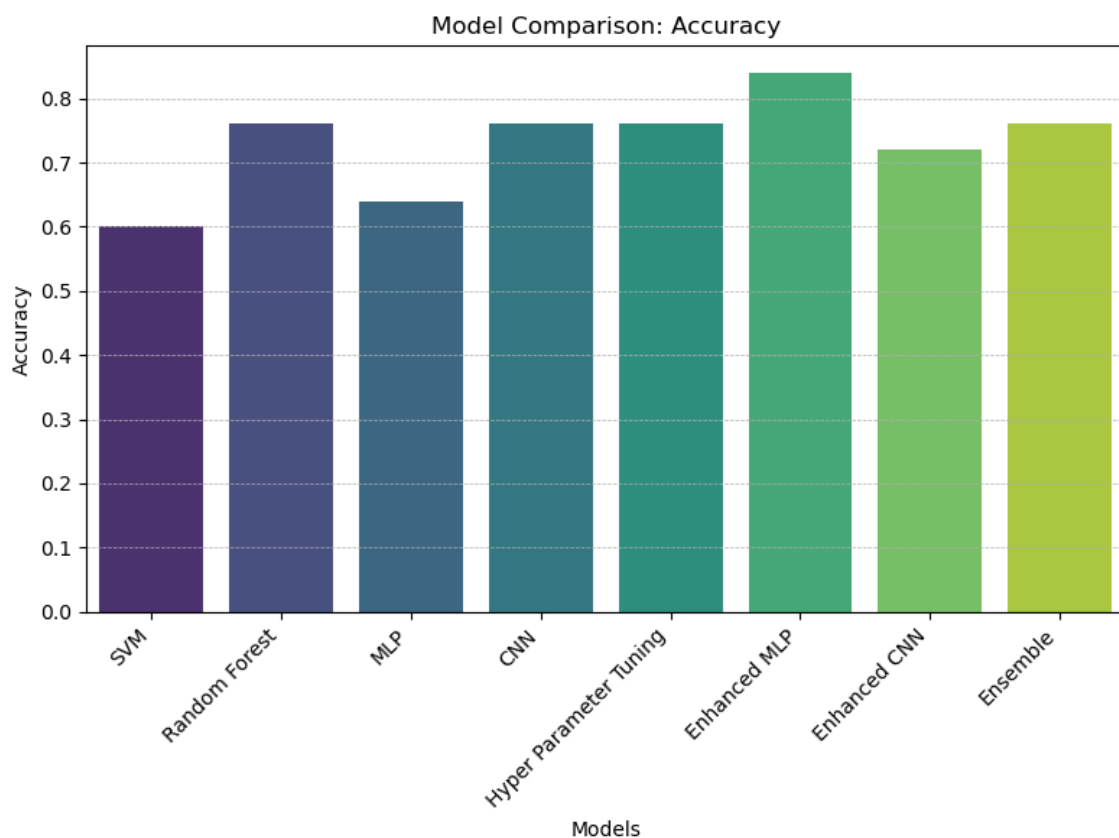
Features

Category	Feature	Description
Low-Level Descriptors (LLDs)	MDVP:Fo(Hz)	Average vocal fundamental frequency (pitch).
	MDVP:Fhi(Hz)	Maximum vocal fundamental frequency.
	MDVP:Flo(Hz)	Minimum vocal fundamental frequency.
	MDVP:Jitter(%)	Variation in fundamental frequency (pitch stability).
	MDVP:Jitter(Abs)	Absolute variation in fundamental frequency.
	MDVP:RAP	Relative average perturbation in pitch.
	MDVP:PPQ	Pitch period perturbation quotient.
	Jitter:DDP	Average absolute difference of differences in pitch periods.
	MDVP:Shimmer	Amplitude variation of the voice signal.
	MDVP:Shimmer(dB)	Amplitude variation in decibels.
	Shimmer:APQ3	Amplitude perturbation quotient for 3 periods.
	Shimmer:APQ5	Amplitude perturbation quotient for 5 periods.
	MDVP:APQ	Amplitude perturbation quotient.
	Shimmer:DDA	Average absolute differences in amplitude perturbations.
	NHR	Noise-to-harmonics ratio.
	HNR	Harmonics-to-noise ratio.
Prosodic Features	Spread1	Nonlinear measure of fundamental frequency variation.
	Spread2	Another nonlinear measure of frequency variation.
	PPE	Pitch period entropy (signal irregularity).
Voice Quality Features	NHR	Measures the ratio of noise to tonal components in the voice.
	HNR	Measures the tonal-to-noise quality of the voice.
Temporal Features	Status	Binary classification (0 = Healthy, 1 = Parkinson's Disease).

Feature set after extraction

1	label	file_id	P_sma_de	ma_de_up	sma_de	NR_sma_lp	sma_de	d250-650	pand250-6	and250-650	ncropped	al_sma3nz	sma3nz_std	dB_sma3nz	CF_sma3nz	sma3nz_sm	27.5Hz_s	7.5Hz_sma3	5Hz_sma3
2	0	File_1	0.000119	0.997214	0.994429	36.76051	0.013368	0.730759	0.270677	0.905557	0.810072	0.004371	0.52941	0.603546	8.307067	0.079763	27.18283	0.015623	2.821403
3	0	File_2	1.63E-05	0.995238	0.985782	60.27857	0.020692	0.719499	0.488884	0.370838	0.768965	0.005793	0.610212	0.523561	6.499732	0.126182	24.14716	0.00958	3.413859
4	0	File_3	0.0001	0.996324	0.992647	45.481	0.020426	0.534799	0.544552	1.494704	0.84041	0.004847	1.185104	0.940736	13.0186	0.110631	37.13166	0.00326	2.39556
5	0	File_4	6.46E-05	0.992701	0.989091	47.53808	0.010462	0.753957	0.220333	0.589418	0.762617	0.004663	1.517298	0.553196	6.399995	0.15903	22.87894	0.00803	2.658834
6	0	File_5	8.56E-06	0.99511	0.97561	2.086884	0.130255	0.878799	0.099037	0.082972	0.771283	0.004141	0.571693	0.556314	6.188959	0.130538	24.28415	0.008939	4.338348
7	0	File_6	0.000188	0.996875	0.996885	39.44236	0.020801	0.641419	0.161547	0.562056	0.783358	0.004229	0.818656	0.545755	7.05082	0.120728	24.72936	0.006083	1.868297
8	0	File_7	1.34E-05	0.991525	0.984	59.41971	0.047398	0.800872	0.526913	0.85119	0.720178	0.008696	0.788819	0.644453	5.080616	0.23658	22.45308	0.013494	9.178717
9	0	File_8	4.86E-05	0.997519	0.990123	16.74373	0.028888	0.649373	0.038318	0.032612	0.849622	0.001848	0.886096	0.547565	9.498416	0.17832	33.35426	0.004924	2.251751
10	0	File_9	0.000904	0.951613	0.183206	166.996	0.090252	0.817565	0.146393	0.074531	0.727875	0.078515	0.982947	0.432507	2.182307	1.40611	29.04299	0.17235	91.86913
11	0	File_10	5E-06	0.997525	0.007389	30.45951	0.016359	0.520198	0.250691	0.573884	0.766262	0.002844	0.488474	0.715603	6.091091	0.064152	22.13969	0.005644	2.501319
12	0	File_11	6.14E-05	0.996678	0.990066	44.72559	0.034291	0.503657	0.068509	0.315483	0.839147	0.004718	1.43806	0.491109	10.38769	0.132731	33.12613	0.003678	2.646202
13	0	File_12	1.45E-05	0.973118	0.090426	5.58499	0.285874	0.750263	0.325891	1.395719	0.817325	0.00511	0.748255	0.519714	10.33454	0.119012	32.76296	0.006991	4.035458
14	0	File_13	0.000188	0.997555	0.992736	34.69785	0.012215	0.862869	0.192216	0.220872	0.791674	0.023427	0.837929	0.457139	5.479779	0.56967	36.97273	0.021806	18.66579
15	0	File_14	0.000246	0.996248	0.988785	1.028153	0.167819	0.846244	0.42895	0.225566	0.834676	0.013784	3.493361	0.490323	7.839237	0.34549	35.09814	0.067477	37.58937
16	0	File_15	0.000213	0.993151	0.013514	76.23044	0.023961	0.617722	0.251401	0.356447	0.83928	0.004778	0.572309	0.570625	10.0366	0.18711	35.6488	0.009786	0.511265
17	0	File_16	8.77E-05	0.997319	0.97861	18.9733	0.018904	0.593785	0.249299	0.334607	0.826916	0.003266	0.821073	0.750238	10.60826	0.106428	32.19827	0.005261	2.673727
18	0	File_17	0.00082	0.986755	0.980263	78.02596	0.026648	0.809338	0.11856	0.333884	0.80396	0.005497	4.28397	1.106651	8.303356	0.249542	28.32161	0.066619	12.42086
19	0	File_18	6.06E-05	0.983333	0.006515	35.85249	0.230841	0.772157	0.625021	0.614282	0.737413	0.014922	2.486337	0.660664	4.500729	0.395132	22.86854	0.085096	17.63007
20	0	File_19	5.43E-06	0.972851	0.991031	4.958771	0.245774	0.812958	0.421866	0.317666	0.836626	0.004107	0.54898	0.550134	11.6278	0.092665	34.37701	0.005387	2.56212
21	0	File_20	4.85E-05	0.997375	0.018277	3.384665	0.294775	0.818057	0.098862	0.277868	0.775518	0.004306	0.804775	0.55224	6.700254	0.144088	25.07414	0.012516	4.331388
22	0	File_21	0.000216	0.987342	0.975	5.955611	0.156743	0.642571	0.453665	0.703595	0.82296	0.003044	1.526034	0.792992	9.343796	0.107905	29.01451	0.009989	0.993865
23	0	File_22	0.00014	0.98374	0.977444	185.0956	0.066155	0.643353	0.755287	0.672595	0.736772	0.028534	1.13022	0.912184	5.237359	0.832558	29.77162	0.115806	6.683777
24	0	File_23	1.05E-05	0.988095	0.011765	49.81573	0.028086	0.814028	0.564376	0.841092	0.769338	0.004868	0.624273	0.508443	6.011763	0.182311	25.1987	0.008513	2.337513
25	0	File_24	0.000164	0.99373	0.990625	22.27374	0.027469	0.44629	0.130951	0.114155	0.859768	0.002203	1.465037	0.559201	11.24032	0.144839	35.4206	0.004923	1.911432
26	0	File_25	7.21E-05	0.996	0.984064	24.54577	0.015938	0.821259	0.375839	0.15854	0.809133	0.005063	0.580259	0.511123	10.7352	0.075736	33.32002	0.004728	1.343184

Model Comparison:



	Model	Accuracy	Precision	Recall	F1-Score
0	SVM	0.60	0.500000	0.3	0.375000
1	Random Forest	0.76	0.642857	0.9	0.750000
2	MLP	0.64	0.545455	0.6	0.571429
3	CNN	0.76	0.700000	0.7	0.700000
4	Hyper Parameter Tuning	0.76	0.642857	0.9	0.750000
5	Enhanced MLP	0.84	0.800000	0.8	0.800000
6	Enhanced CNN	0.72	0.636364	0.7	0.666667
7	Ensemble	0.76	0.666667	0.8	0.727273

Literature Study

A machine learning method to process voice samples for identification of Parkinson's disease

Objective:

- Explore the reliability of low-quality voice recordings collected via telephone for identifying Parkinson's Disease (PD) using machine learning (ML) and deep learning (DL) approaches.

Innovations:

- Novel use of Inception V3 CNN with transfer learning for spectrogram classification.
- Analysis of spectrogram images retains temporal and frequency energy distribution of the sustained vowel sound (/a/).
- Comparison of deep learning results with traditional statistical classifiers (Random Forest, Logistic Regression).

Key Results:

- CNN-based spectrogram analysis achieved **AUC = 0.97** for colored spectrograms and **AUC = 0.96** for grayscale spectrograms, outperforming traditional methods.
- Demonstrated feasibility of using low-quality recordings (8 kHz) for PD detection, supporting clinical applications in resource-limited areas.

Impact:

- Simplifies PD diagnosis via non-invasive, accessible, and remote voice-based screening, enabling earlier detection and rural inclusivity.



Adapted Methodologies

1) Data Collection and Preprocessing:

- Voice recordings were collected via telephone voicemail at an 8 kHz sampling rate.
- Background noise removal using decibel-based filters (75–300 dB for males, 100–600 dB for females).
- Trimmed silent segments using short-time energy thresholds.
- Result: Processed 1.5-second sustained vowel /a/ samples.

2) Acoustic Signal Features Extraction:

- Used Opensmile (Python) interface.

Extracted 23 features:

- Fundamental frequency (F0) – mean and standard deviation.
- Harmonics-to-noise ratio (HNR).
- Jitter (frequency variations) and shimmer (amplitude variations).
- Formant frequencies (F1–F4).

Classified using Random Forest and Logistic Regression.



Adapted Methodologies

3. Spectrogram Creation:

- Spectrograms generated using Fast Fourier Transform (FFT) with:
 - 32 ms sliding window, 50% overlap, 1024 FFT size.
 - Represented signal energy across time and frequency axes.
- Grayscale and colored spectrograms were created and normalized.

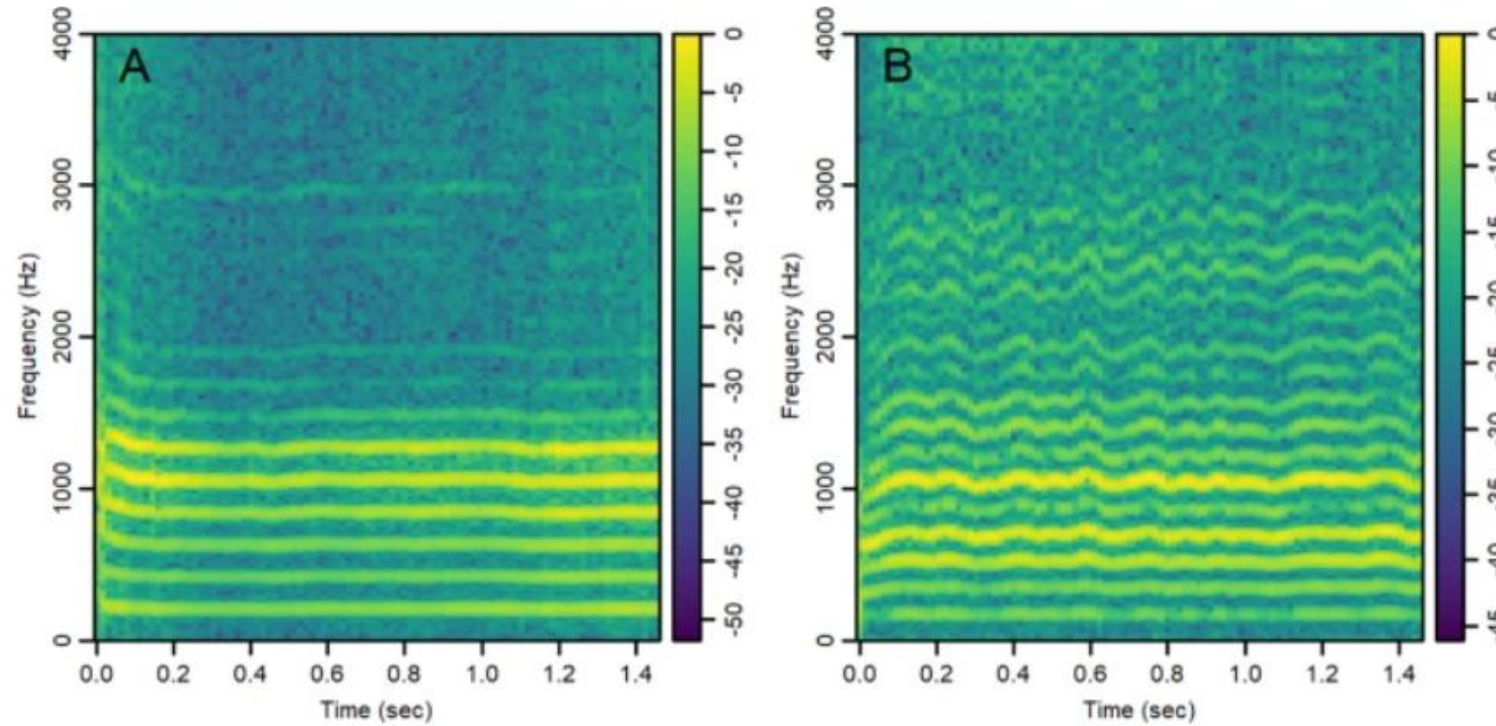
4. Convolutional Neural Network (CNN):

- Pre-trained Inception V3 CNN architecture (transfer learning):
 - Trained on ImageNet database; adapted to classify spectrograms into HC and PwPD.
- Fine-tuned with 4 additional layers:
 - Batch normalization, Dense, Dropout, and Final Dense layers.
- Trained with Adam optimizer, 0.001 learning rate, 10 epochs, batch size = 4.
- Achieved superior classification with AUC = 0.80 for color spectrograms.

5. Validation and Comparison:

- Cross-validation with 70% training and 30% testing in 100 random splits.
- Comparison with RF and LR classifiers using acoustic and spectral features.
- CNN showed better robustness and smaller variability across iterations.





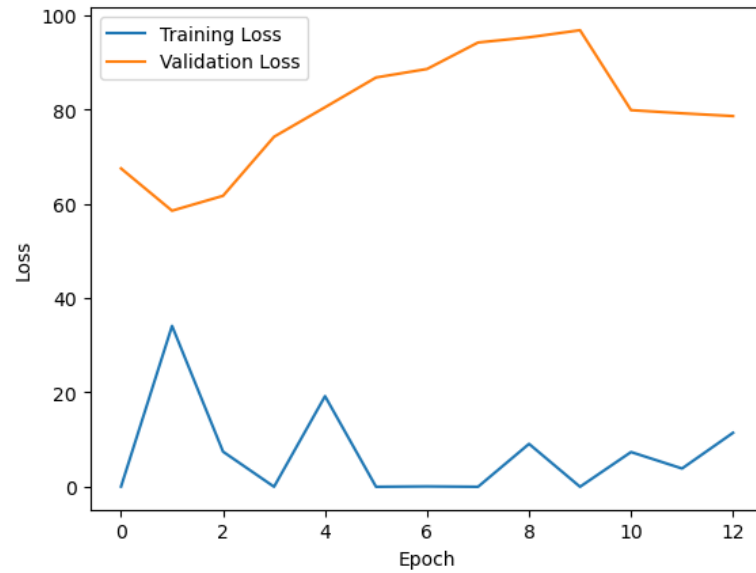
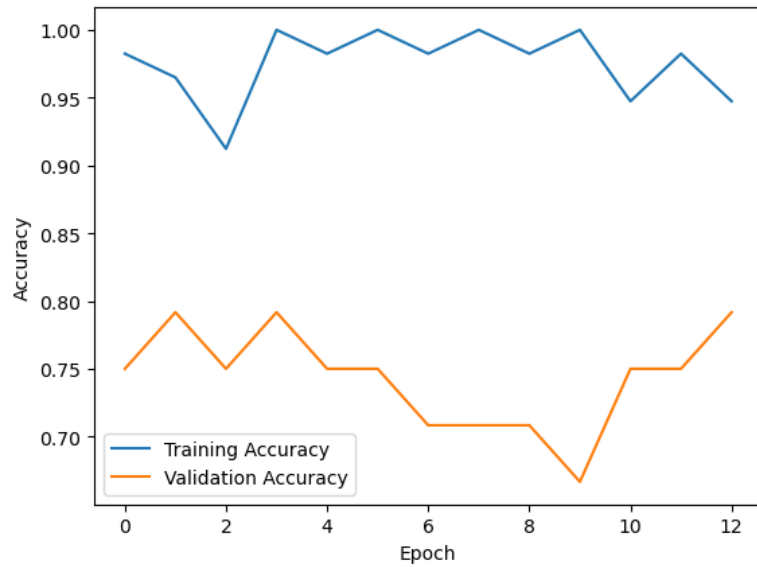
Spectrogram Analysis

Colored spectrograms of 1.5 s of the sustained vowel /a/ uttered by selected subjects: **(A)** Healthy control (female), **(B)** Parkinson's patient (male). The color scale represents $10 \log_{10}(|S|/\max(|S|))$, where S represents the complex numbers at the output of the FFT.

	Healthy controls (n = 41)	Parkinson's disease (n = 40)
Sex (male/female)	16/24	21/19
Age at enrollment (years)	47.9 ± 14.5	66.6 ± 9.0
Hoehn & Yahr stage of PD	–	2.1 ± 0.4
Disease duration (years)	–	9.5 ± 6.0



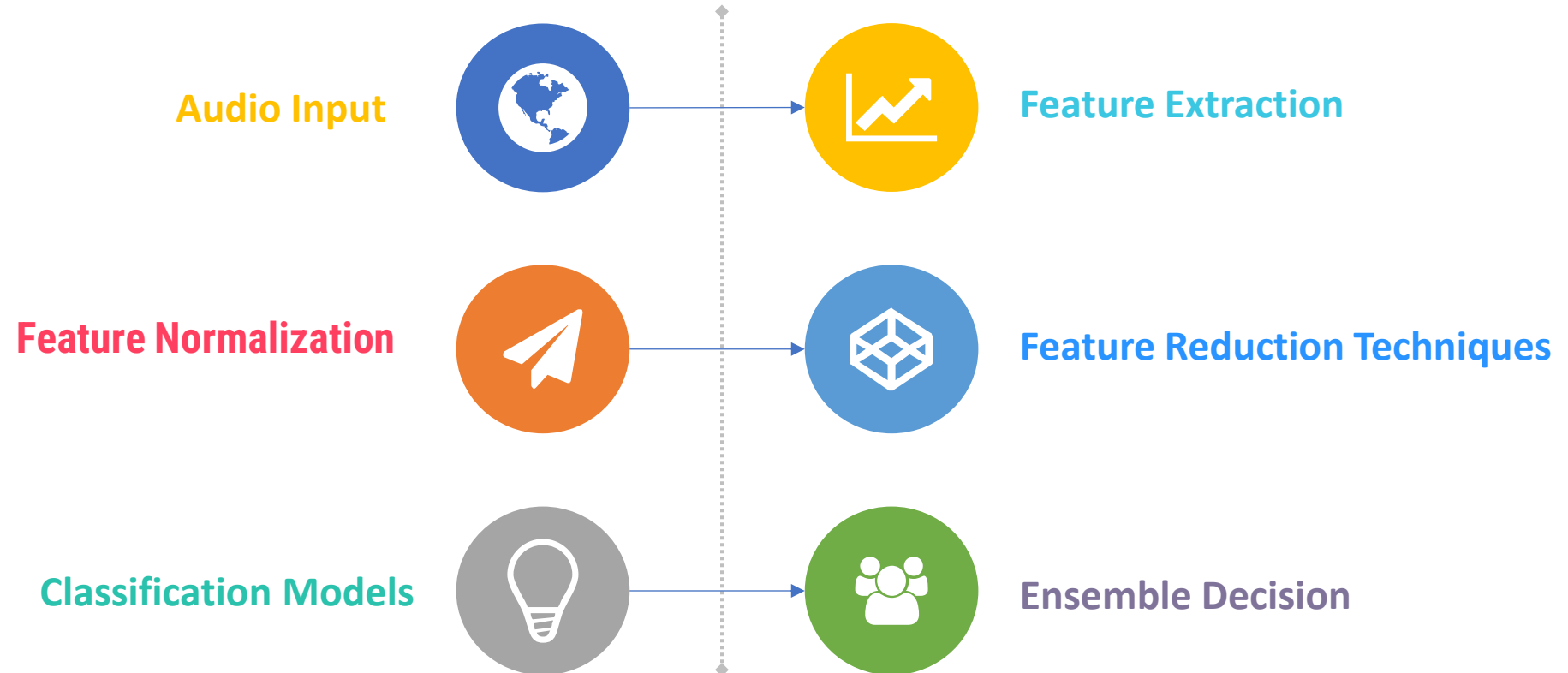
Dataset Analysis



Achieved 80% Accuracy

**Model
Performance**

Our Methodology



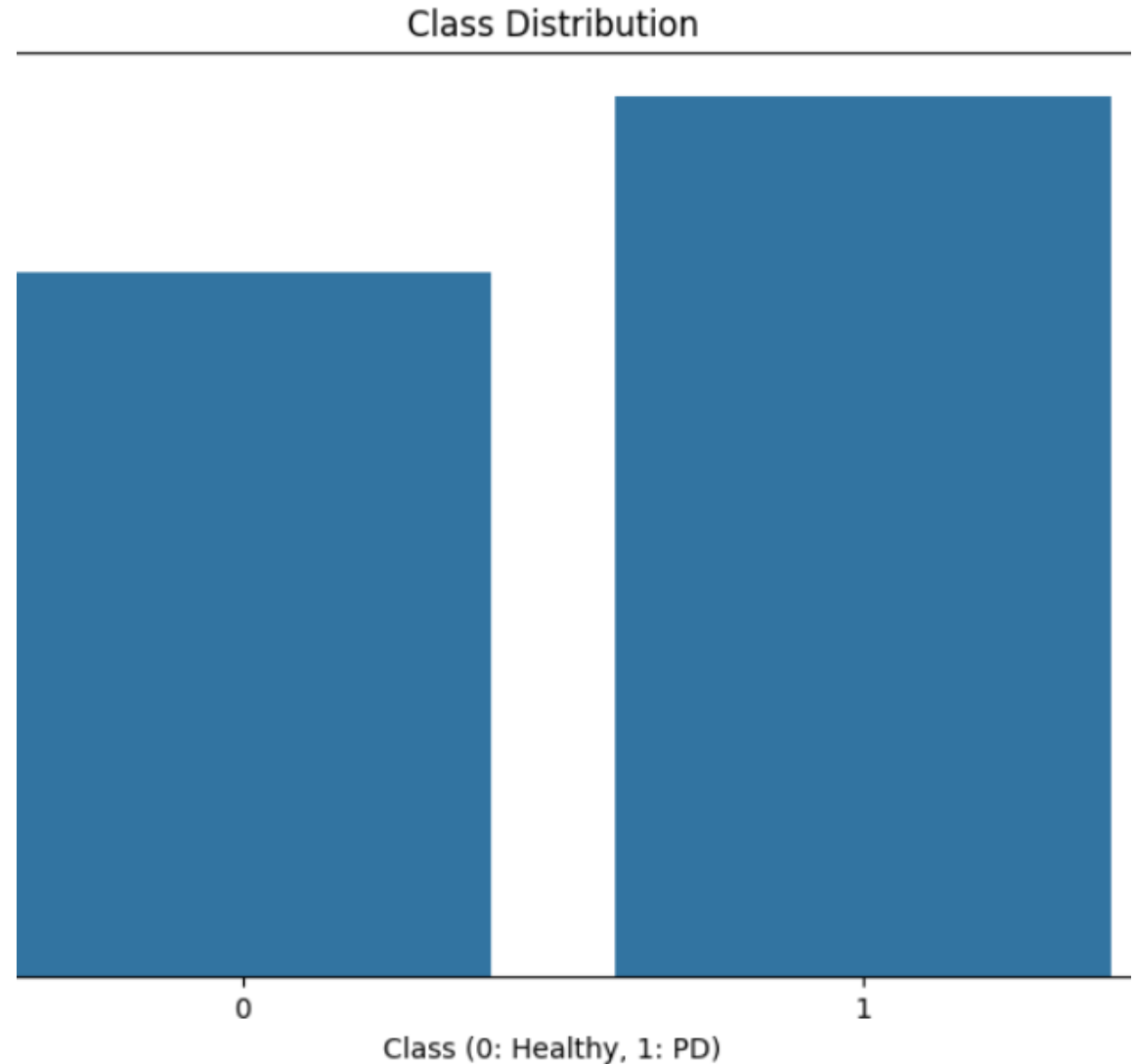
Italian Dataset Analysis

Young Healthy Controls (YHC): 15 individuals (13 males, 2 females), aged 19–29.

Healthy Elderly Controls (HEC): 22 individuals (10 males, 12 females), aged 60–77.

Parkinson's Disease (PD) Patients: 28 individuals (19 males, 9 females), aged 40–80

- Number of healthy control samples: 220
- Number of PD samples: 275
- Processed only the “AAAh” sounds which are in different patterns
- Number of features Extracted: 88



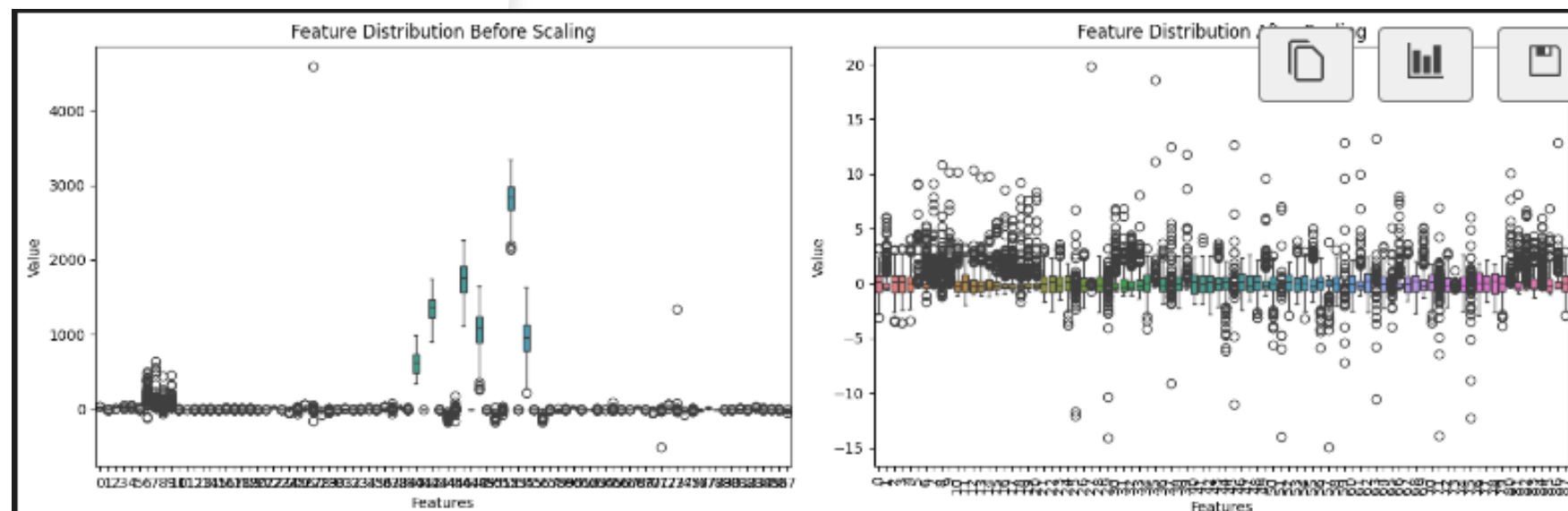
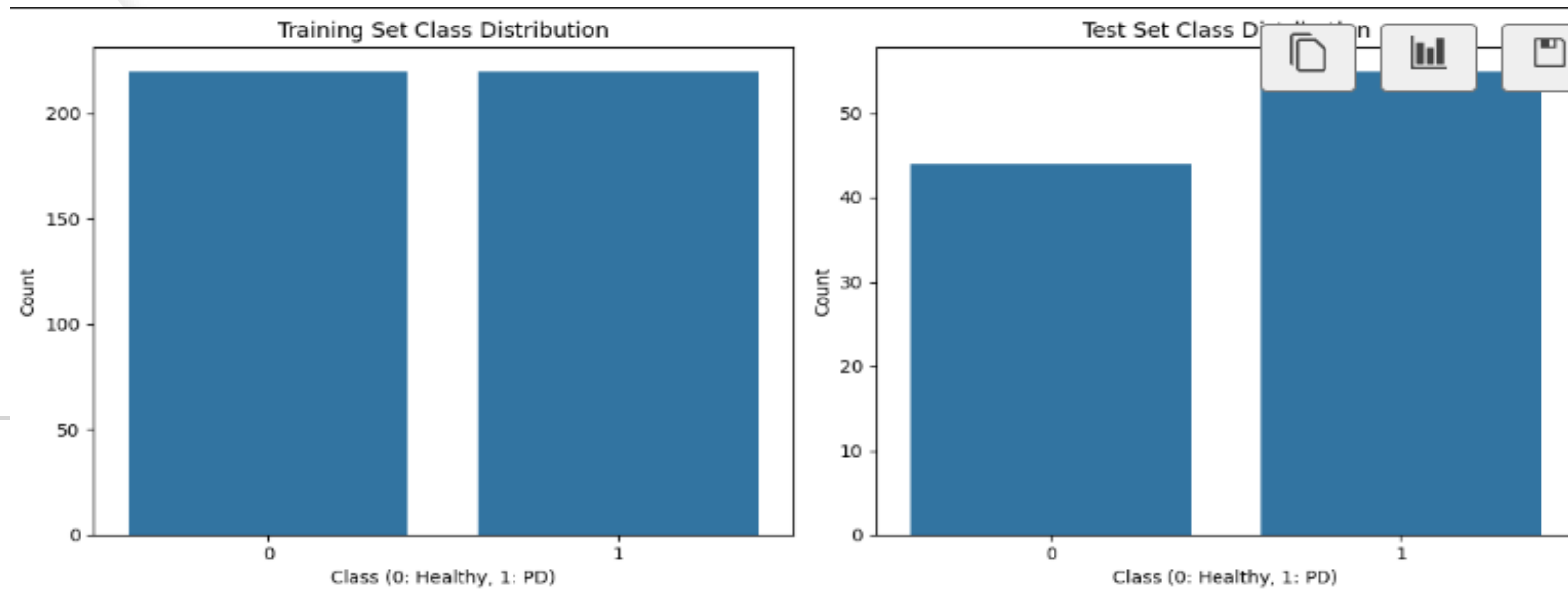


Feature Extraction

OpenSmile eGeMAPS v02 Features:

- Frequency-related:
 - F0 (pitch)
 - Jitter
 - Formant frequencies (F1, F2, F3)
- Energy/Amplitude-related:
 - Shimmer
 - Loudness
 - Harmonics-to-Noise Ratio
- Spectral:
 - MFCC
 - Alpha Ratio
 - Hammarberg Index

- StandardScaler application
- Robust to outliers



Machine Learning Pipeline

Feature Reduction Techniques:

- PCA (20 components)
- SelectKBest (20 features)

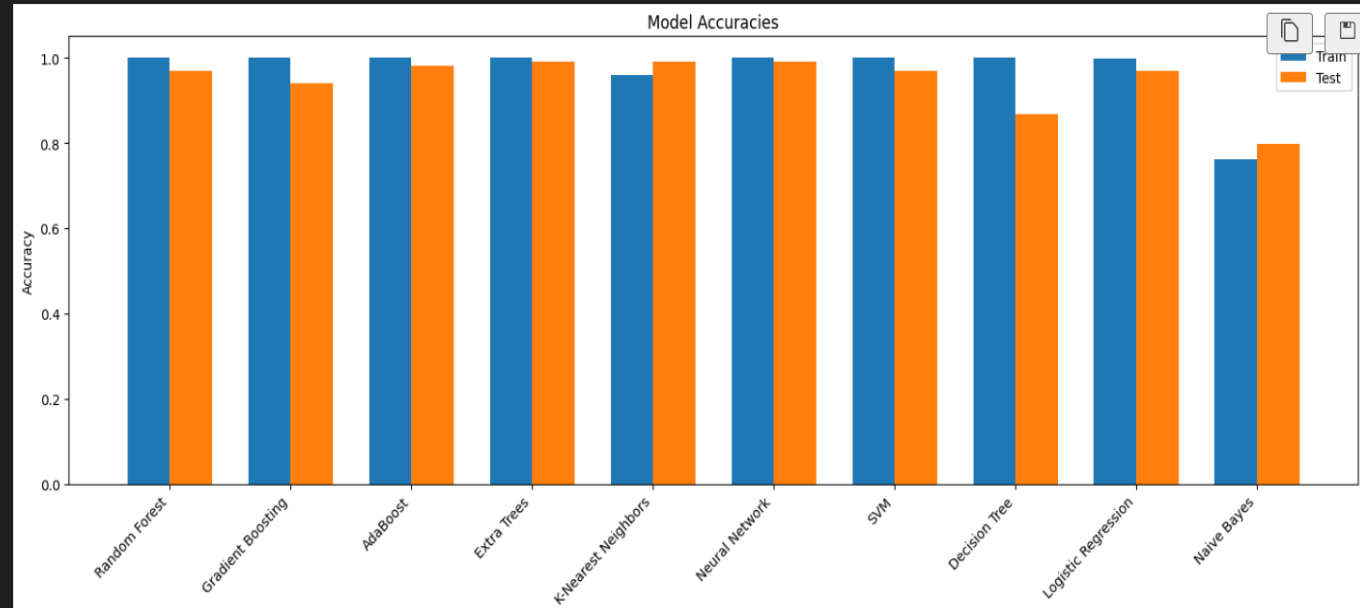
Classification Models:

- Random Forest
- Gradient Boosting
- SVM
- Neural Network
- KNN
- Decision tree
- Naïve Bayes
- Logistic Regression

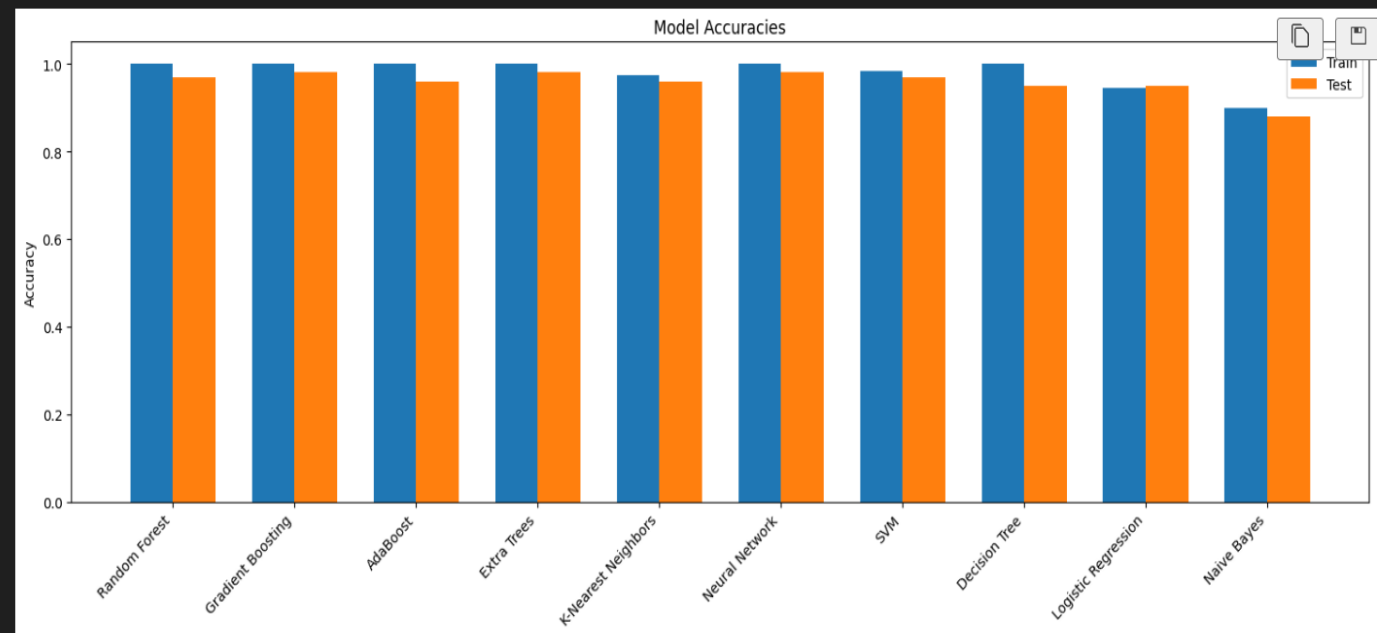
Model Evaluation

- Training Results
- Cross-validation Scores
- Performance Metrics

PCA Results:



SelectKBest Results:



Real-time Classification

- 1) Recording Interface
- 2) Feature Visualization
- 3) Multi-model Predictions
- 4) Consensus Decision

Parkinson's Disease Voice Classification

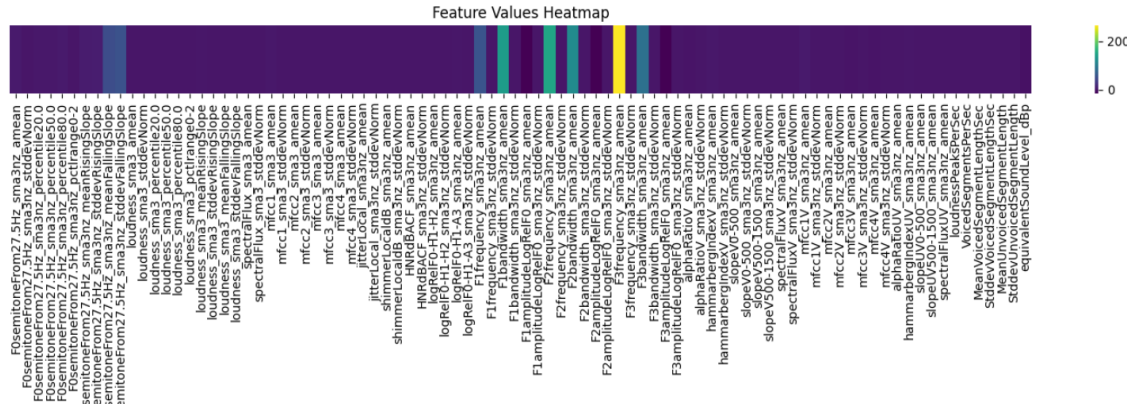
Click to record your voice for 5 seconds.

🎤 Record (5 sec)

🎵 Your Recording:

▶ 0:00 / 0:05 🔊 ⋮

📊 Extracted Features:



Key Features (first 5):

F0semitoneFrom27.5Hz_sma3nz_amean	28.301809
F0semitoneFrom27.5Hz_sma3nz_stddevNorm	0.105586
F0semitoneFrom27.5Hz_sma3nz_percentile20.0	27.030701
F0semitoneFrom27.5Hz_sma3nz_percentile50.0	27.191715
F0semitoneFrom27.5Hz_sma3nz_percentile80.0	32.164013



Classification Results:

Overall Consensus:

Healthy (Agreement: 62.5%)

Average Confidence: 92.3%

SelectKBest - KNN

Prediction: Healthy

Confidence: 100.00%

SelectKBest - Decision Tree

Prediction: Healthy

Confidence: 100.00%

PCA - KNN

Prediction: Healthy

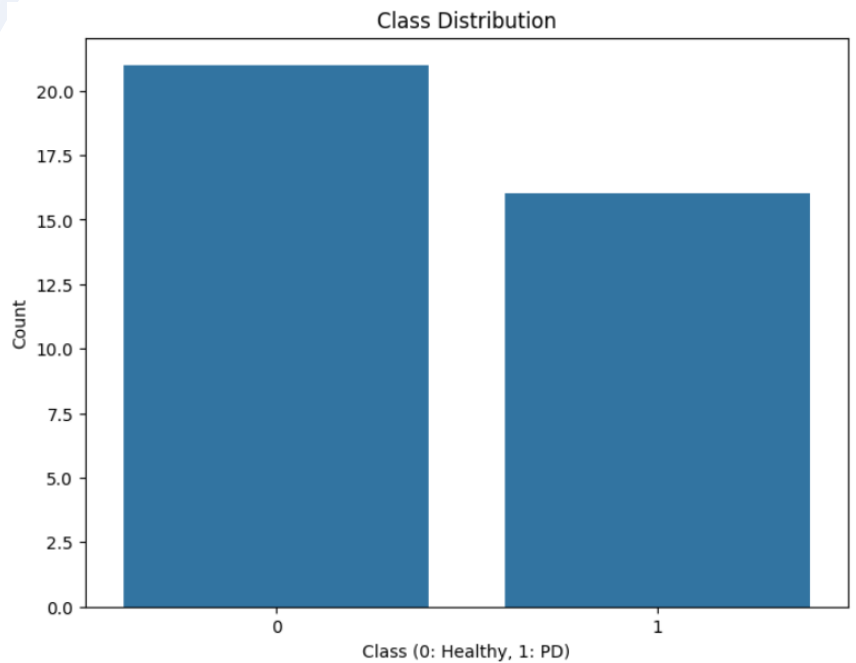
Confidence: 100.00%

PCA - Decision Tree

Prediction: Healthy

Confidence: 100.00%

KCL Parkinson's Dataset Analysis



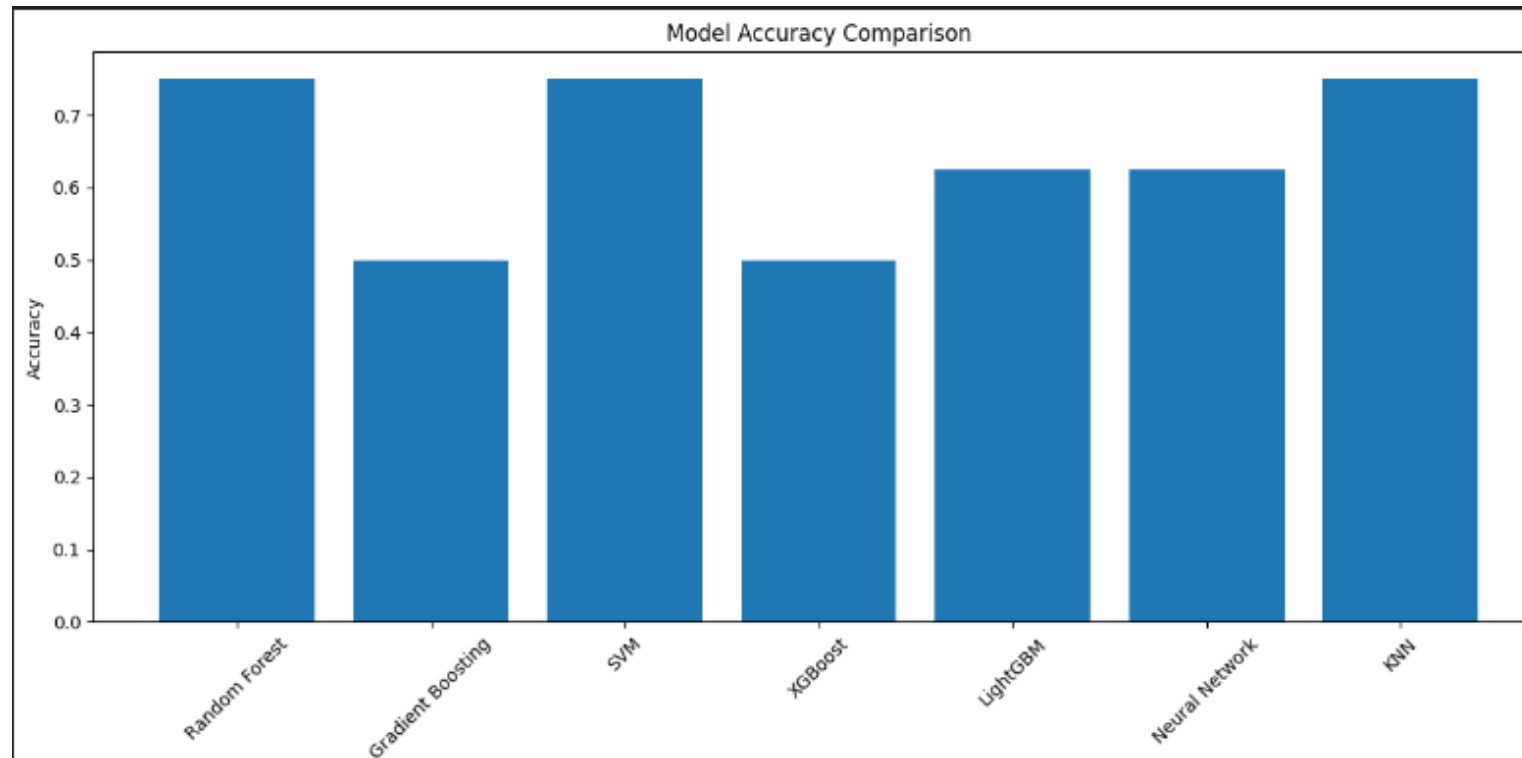
Parkinson's Disease (PD) Patients: X individuals (male/female distribution), with varying disease severity.

Healthy Controls (HC): X individuals (male/female distribution),

- Large dataset includes varied speech tasks for comprehensive analysis.
- Has the Both Readtext and SpontaneousDialogue.
- Number of healthy control samples: 21
- Number of PD samples: 16

Model Comparison:

RandomForest, KNN, SVM are best performing Algorithms with 75% accuracy



Future Scope



Development of Multilingual Models

Build language-independent frameworks to accommodate diverse linguistic populations.



Combining Multimodal Data

Integrate speech features with other biomarkers like gait and handwriting for comprehensive diagnostics.



Early Diagnosis Enhancement

Develop systems to identify pre-symptomatic markers, aiding in early intervention strategies.

Real-Time Monitoring Tools

Create mobile and wearable applications for continuous speech-based monitoring of PD progression..



Integration with Telemedicine

Incorporate speech analysis systems into telehealth platforms for remote and cost-effective diagnosis..



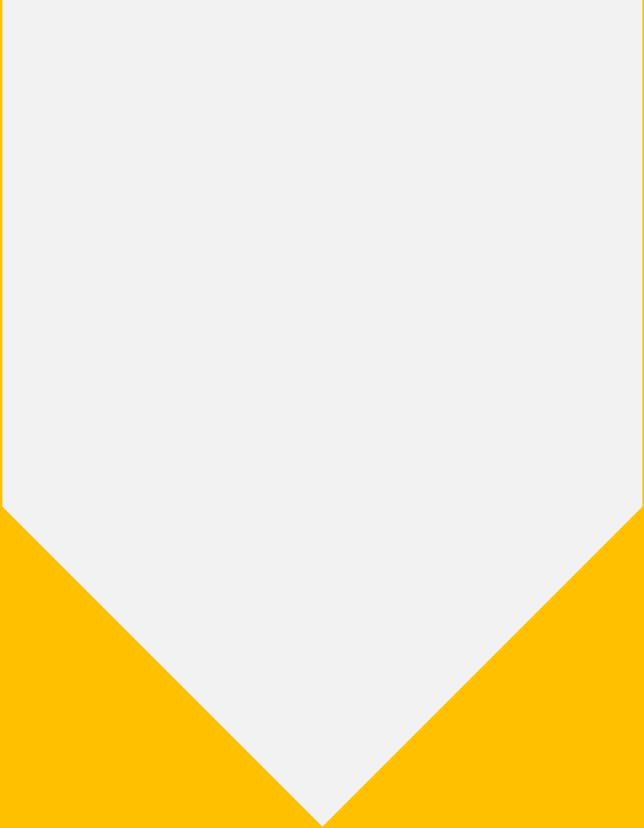
Personalized Treatment Plans

Use speech data to track individual disease progression and tailor therapeutic approaches



References

- Italian Parkinson dataset : Italian Parkinson's Voice and Speech | IEEE DataPort
- KCL dataset :Mobile Device Voice Recordings at King's College London (MDVR-KCL) from both early and advanced Parkinson's disease patients and healthy controls
- Comparative Study of Speech Analysis Methods to Predict Parkinson's Disease2111.10207
- Voice Analysis for Neurological Disorder Recognition—A Systematic Review and Perspective on Emerging Trends<https://doi.org/10.3389/fdgth.2022.842301>
- Automated Assessment of Psychiatric Disorders Using Speech: A Systematic Review Preprints | Automated Assessment of Psychiatric Disorders Using Speech: A Systematic Review



THANK YOU !

